



LLM Agents and their Ability to Support Feature Construction in Data Mining Tasks

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1 Introduction

Feature engineering plays a vital role in the success of machine learning models, as the quality of features directly impacts model performance and interpretability. While existing automated feature generation methods have improved efficiency, they often lack explainability, flexibility, and the ability to generalize across different data types and tasks.

Recent advances in large language models (LLMs) offer new possibilities for dynamic and context-aware feature generation through prompt-based approaches. Building on this idea, this project aims to enhance LLM-driven feature generation by making it more adaptive, interpretable, and applicable to real-world scenarios.

The work focuses on improving prompt design, ensuring consistency in generated features, and enabling integration into practical data pipelines. Ultimately, the goal is to develop a more flexible and reliable feature generation framework that bridges the gap between automated methods and human understanding.

Furthermore, the framework will be evaluated across diverse datasets to assess its robustness and scalability in different domains. This research also aims to contribute towards more transparent and trustworthy AI systems by aligning automated feature engineering with human interpretability and domain knowledge.

2 Project Architecture

The project follows a prompt-driven feature generation strategy using large language models (LLMs). First, raw data is collected and preprocessed through standard ETL steps such as cleaning, normalization, and schema alignment. Then, carefully designed prompts are used to guide the LLM in generating meaningful, domain-aware features from the input data.

An iterative refinement process is applied where generated features are evaluated using statistical methods and model performance metrics. Based on this feedback, prompts are adjusted to improve feature quality, consistency, and relevance. Additionally, a human-in-the-loop component can be incorporated to validate and interpret generated features, ensuring reliability and explainability.

Finally, the selected features are integrated into machine learning models, and their impact is measured against baseline feature engineering techniques.

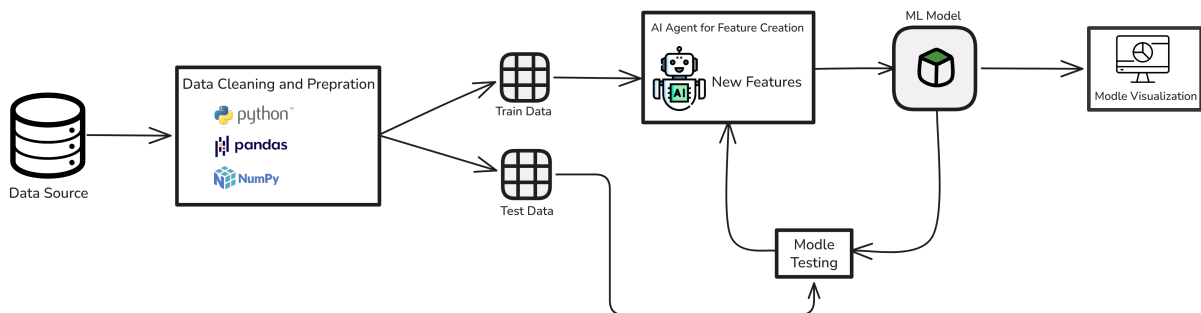


Figure 1: Project Architecture Diagram

3 Scientific Relevance for Artificial Intelligence

The emergence of large language models (LLMs) has significantly influenced Artificial Intelligence, particularly in knowledge representation and utilization. This project contributes to AI research by exploring how LLM-based agents can automate and enhance feature construction, a critical step in data mining and machine learning workflows.

Traditionally, feature construction relies on domain expertise and manual effort. By leveraging LLM agents, this project introduces a new paradigm where semantic understanding and contextual reasoning are used to generate meaningful, domain-aware features. This shifts feature engineering toward a more adaptive and scalable approach.

From a scientific perspective, the project investigates how prompt engineering and agent-based frameworks enable LLMs to perform structured analytical tasks beyond natural language processing. It examines their ability to generalize across domains and improve model performance through intelligent feature generation.

Additionally, incorporating iterative feedback loops and human-in-the-loop validation supports the development of reliable and explainable AI systems. Overall, this work positions LLM agents as active contributors to the data science pipeline, enhancing efficiency and supporting the development of more intelligent and human-aligned AI systems.

4 Planning

The development of the project “*LLM Agents and their Ability to Support Feature Construction in Data Mining Tasks*” follows a structured and iterative approach to ensure both scientific rigor and practical effectiveness. The planning is divided into the following key phases:

- **Problem Definition and Objective Setting**

Clearly define the research problem and objectives. This includes identifying the limitations of traditional feature engineering methods and formulating research questions on how LLM agents can enhance or automate feature construction.

- **Data Collection and Understanding**

Select relevant datasets based on the domain. Perform Exploratory Data Analysis (EDA) to understand data distributions, missing values, correlations, and potential challenges.

- **Data Preprocessing and ETL Pipeline**

Clean, transform, and standardize data using tools such as Python, Pandas, and NumPy. Tasks include handling missing values, normalization, encoding categorical variables, and splitting the dataset into training and testing sets.

- **Baseline Feature Engineering**

Develop a baseline model using conventional feature engineering techniques to serve as a benchmark for evaluating LLM-generated features.

- **Design of LLM Agent and Prompt Strategy**

Design an LLM-based agent and create carefully crafted prompts to guide the model in generating meaningful and context-aware feature transformations.

- **Feature Generation and Integration**

Generate candidate features using the LLM agent. Validate and integrate these features into the dataset using statistical methods and optional human validation.

- **Model Training and Evaluation**

Train machine learning models using both baseline and enhanced feature sets. Evaluate performance using metrics such as accuracy, precision, recall, F1-score, or RMSE.

- **Iterative Refinement**

Refine prompts and feature generation strategies based on evaluation results to improve feature quality and model performance.

- **Visualization and Interpretation**

Visualize results using tools such as Power BI or Python libraries. Analyze feature importance and model interpretability.

- **Thesis Documentation and Reporting**

Document all methodologies, experiments, and findings. Highlight the effectiveness, limitations, and future potential of LLM agents in feature engineering.

5 Resources and Support

The project will use a standard data science and machine learning stack, including Python and key libraries such as Pandas, NumPy, and Scikit-learn for data processing, feature engineering, analysis, and model development. For implementing LLM-based agents, access to large language model APIs or locally hosted models will be required, depending on availability and computational constraints.

The implementation can be carried out on a personal computer with sufficient processing capacity; however, the use of cloud computing platforms or high-performance computing resources can significantly enhance performance, particularly when working with large datasets, complex models, or multiple experimental runs.

For data exploration, analysis, and visualization, tools such as Jupyter Notebooks and Power BI will be used to support interactive development, result interpretation, and clear presentation of insights. GitHub will be utilized for version control, code management, reproducibility, and collaborative development.

Academic supervision, continuous feedback, and access to relevant scientific literature will guide the project throughout its lifecycle. Research papers, technical documentation, and existing work in areas such as large language models, feature engineering, and data mining will ensure the project remains aligned with current academic research and industry best practices.

References

- X. Zhang, J. Zhang, B. Rekabdar, Y. Zhou, P. Wang, and K. Liu, “Dynamic and adaptive feature generation with llm,” *arXiv preprint arXiv:2406.03505*, 2026.
- N. Hollmann, S. Müller, and F. Hutter, “Large language models for automated data science: Introducing caafe for context-aware automated feature engineering,” *arXiv preprint arXiv:2305.03403*, 2023.